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14 INITIAL TESTS AND OPERATION

14.1 Description of Test Program

A carefully conceived and executed startup testing program, under the control, responsibility, and authority of Public Service Electric and Gas, was implemented to accomplish a safe, orderly, and comprehensive startup. This program demonstrated that the plant operates satisfactorily and presents no danger to the health and safety of the public. The program also provided an opportunity for station personnel to become thoroughly familiar with plant systems and procedures and to determine the adequacy of normal and emergency plant operating procedures.

After installation, individual components and systems were tested and evaluated in accordance with approved written test procedures. Although these preoperational test procedures were written for construction type testing, references to the normal operating and test procedures contained in the Plant Manual were made where feasible to assist in verification of their accuracy and applicability. Analyses of test results were made to verify that systems and components performed satisfactorily and to determine corrective action, if required.

The program included tests, adjustments, calibrations, and system operations necessary to assure that initial fuel loading, initial criticality and subsequent power operation could be safely undertaken. In general, tests were classified by type, as integrity, functional, operational, hydrostatic, cleaning, etc. Integrity tests verified proper installation of equipment. Functional tests verified that the systems or equipment were capable of performing the function for which they were designed. Operational tests involved actual operation of the systems and equipment to demonstrate the state of readiness and capability of the systems. Whenever feasible, tests were performed under conditions similar to normal station operation. During system tests for which unit parameters were not available and could not be simulated, systems were operationally tested as far as possible without these parameters.

The remainder of the tests were performed as the parameters became available. Abnormal unit conditions were simulated during testing as required, whenever such conditions did not endanger personnel or equipment, or contaminate clean systems.

Prior to participating in preoperational tests, station operators and supervisors completed extensive formal classroom training in reactor theory and plant systems. In addition, considerable on-the-job training was obtained while verifying systems' completion prior to initiating tests.

The test program consisted of three distinct phases as described below:

Phase I

COMPONENT PREOPERATIONAL TESTS – Post-construction testing to verify component functional characteristics and installation integrity and confirmation of readiness for system operational tests. Tests included instrument calibration, continuity and megger checks, motor bump, hydrostatic testing, cleaning and flushing.

Phase II

SYSTEM OPERATIONAL TESTS – Individual system tests to verify satisfactory system operation and performance in order to ensure that systems could safely support initial criticality and subsequent testing at power.

Phase III

PLANT OPERATIONAL TESTS – Nuclear operation of the reactor, beginning with initial core load and including all power ascension and plant acceptance tests. Tests were performed to establish operating characteristics and ensure conformance with plant license requirements.

14.2 Initial Tests and Operation

14.2.1 Testing Program Description

Prior to initial core loading, a comprehensive testing program was conducted to ensure that all plant systems were properly prepared to support fuel loading and subsequent nuclear testing. To accomplish this, preoperational and operational testing was completed on all systems except for those tests which required the core to be in place. For this reason, operational testing of the Control Rod System and In-core Monitoring System were deferred until after fuel loading.

14.2.2 Test Objectives

In the development of the System Operational Test Program, Public Service Electric & Gas complied with the intent of Regulatory Guide 1.68, dated November 1973. Table 14.2-1 depicts, in their approximate sequence, the tests performed. The program included many nonsafety-related tests and, therefore, exceeded the requirements of the Regulatory Guide.

	System Tests	Test Objectives
1.	Demineralized Water System	To verify quality of water and capacity of system.
2.	Fire Protection System	To verify proper operation of the system by ensuring the design intent is met for pumps, to verify that automatic start functions operate as designed, and to verify that level and pressure controls meet specifications.
3.	Fresh Water System	To verify proper operation of system.
4.	House Heating Boiler Startup	To start up house heating boilers and verify proper operation of house heating steam system.
5.	Heating Water System	To verify proper operation of system.
6.	Compressed Air System	To verify leak tightness of the system, proper operation of all compressors, the manual and automatic operation of controls at design setpoints, design air-dryer cycle time and moisture content of discharge air.
7.	Control Air System	To verify proper operation of system including emergency compressors.
8.	Reactor Coolant System	To verify operation and integrity of system in preparation for hot functional testing.
8.1	Reactor Coolant System Cold Hydrostatic Test	
8.2	Reactor Vessel Internals Measurements	
8.3	Pressurizer Relief Tank	
9.	Hydrostatic Test of the Steam Generator-Secondary Side	To verify integrity of steam generator.
10.	Chemical and Volume Control System	To verify, prior to critical operation, that the Chemical and Volume Control System functions as specified in the FSAR and

	System Tests	Test Objectives
10.1	Volume Control Tank Level and Makeup Control	appropriate manufacturer's technical Manuals. More specifically that:
10.2	Heat Tracing System	
10.3	Boric Acid System	
10.4	Boron Recycle Process	
		(a) All pumps perform to specifications. (b) All temperature, flow, level and pressure controllers function to control at the required setpoint when supplied with appropriate signal(s). (c) All temperature, flow, level and pressure alarms provide alarms at the required locations when the alarm setpoint is reached and clear when the reset point is reached. (d) The reactor makeup control regulates blending, dilution, and boration as designed. (e) The design seal water flow rates are attainable at each reactor coolant pump. (f) Chemical Additional Subsystem functions as specified. (g) Purifications system design flow rates are attainable.
11	Component Cooling System-Cold	To verify component cooling flow to all components and to verify proper operation of instrumentation, controllers, and alarms.
12	Residual Heat Removal System	To verify proper operation of the system.
13	Spent Fuel Pooling Cooling System	To verify proper operation of the system.
14	(Not Used)	

	System Tests	Test Objectives
15	Safety Injection System	To verify that system components and operation are in accordance with requirements as specified in the FSAR and appropriate manufacturer's technical manuals. More specifically that:
15.1	Safety Injection System Performance Test	
15.2	Safety Injection Accumulator Dump Test	
		(a) Safety Injection pumps perform their design functions satisfactorily and reliably. (b) Level and pressure instruments are operable and reset as required. (c) Each pair of valves are installed for redundancy operate as designed. (d) Accumulators can be filled and discharged to the Reactor Coolant System as designed. (e) Flow distribution between injection flow paths are balanced and meet design requirements. (f) Sufficient available centrifugal and safety pumps from NPSH is for the charging injection the RWST.
15	Waste Disposal System	To demonstrate that rated process flow rates can be achieved and that releases can be controlled. To verify proper operation of equipment and systems.
16.1	Liquid Waste Receipt and Storage	
16.2	Liquid Waste Processing -Waste Evaporation	
16.3	Drumming Station	
16.4	Resin Removal System	

System Tests	Test Objectives
16.5 Boiling Station	
16.6 Gaseous Waste Processing	
17. Fuel Handling System	
17.1 Manipulator Crane Indexing	
17.2 Fuel Handling Crane Indexing	To show the system capable of providing a safe and effective means of transporting and handling fuel from the time it reaches the station until it leaves the station. In particular, the tests are designed to verify that:
17.3 Fuel Handling Tools and Fixtures	
17.4 Fuel Transfer System	(a) The major structures required for refueling, such as the reactor cavity, fuel transfer pool, new fuel and spent fuel storage, and decontamination facilities, are in accordance with the design intent. (b) The major equipment required for refueling such as the manipulator crane and fuel handling tools, operate in accordance with the design specifications. (c) All auxiliary equipment and instrumentation function properly. A dummy will be fuel element utilized to conduct these tests.
18. Containment System 18.1 Containment Spray System 18.2 Containment Structural Test	To verify response of the Spray System to control signals and sequencing of the pumps, valves, and controllers as specified in the FSAR and appropriate manufacturer's technical manuals; and to check the time required to actuate the system after a containment high-high

System Tests	Test Objectives
18.3 Containment Leakage Rate Tests (Types A, B, and C)	pressure signal is received. More specifically refer to the test objective listing for the Safety Injection System.
18.4 Containment Isolation System	
18.5 Penetration Cooling System	
	To verify containment integrity by conducting integrated leakage rate and structural integrity tests.
	To verify proper operation of the Isolation and Penetration Cooling Systems.
	To verify proper operability of fans, filters, controls, and other components of 'the Ventilation Systems.
19. Ventilation System	
19.1 Containment Fan Cooler System	
19.2 Iodine Removal System	
19.3 Rod Drive Ventilation System	
19.4 Reactor Nozzle Support Ventilation System	
19.5 Reactor Shield Ventilation System	
19.6 Containment Pressure-Vacuum Relief System	
19.7 Containment Purge System	
19.8 Auxiliary Building Ventilation System	

	System Tests	Test Objectives
19.9	Control Room Air Conditioning System	
19.10	Fuel Handling Building Ventilation System	
20.	NSSS Protection Systems	
20.1	Reactor Protection Time Response Measurement	To verify calibration, operability, and alarm settings of the Reactor Control and Protection System; to test its operability in conjunction with other systems.
20.2	Reactor Protection Operational Check	
20.3	Safeguards System Operational Test	
20.4	Control System Test for Turbine Runback	
20.5	Reactor Plant System Setpoints Verification	
21.	Process Radiation Monitoring System	To verify the calibration, operability, and alarm setpoints of radiation monitors: air particulate monitors, gas monitors, and liquid monitors.
22.	Area Radiation Monitoring System	Same as 21.
23.	Communications System	To verify proper communication between handset stations to ensure proper paging operation, to ensure tie-in between load dispatcher and PA system and adjust amplifiers, speakers and line balance. To verify that all communication stations located at the fuel loading status board locations are functioning.

	System Tests	Test Objectives
24.	Nuclear Instrumentation	The purpose of this test was to verify that the system performs the required indication and control functions through the Source, Intermediate, and Power Ranges of operation. In particular, the tests were designed to verify that: a. The source range instrumentation and protection (high flux level reactor trip) as well as alarm features and audible count rate operate properly. b. The intermediate range instrumentation operates properly, the reactor protective and control features such as high level reactor trip and high level rod stop signals operate properly, and the permissive signals for blocking source range trip and source range high voltage cutoff operate properly. c. The power range instrumentation operates properly, the protective features such as the overpower trips, permissive and dropped rod functions operate with the required redundancy and separation through the associated logic matrices; and the nuclear power signals to other systems are available and operating. d. All auxiliary equipment such as the startup rate channel, recorders and indicators operate properly. e. All instruments are properly calibrated and all set points and alarms are properly adjusted.
25.	Preoperational Testing of Computer	To verify that process computer had been wired properly and that internal converters function properly.

	System Tests	Test Objectives
26.	Computer Input and Data Printout Verification	To verify the operation and calibration of the instrumentation involved in measurement, transmittal, conversion, and printout of process parameters.
27.	Electrical Systems	
27.1	V dc Station Batteries and Charger – Load Test	Energize associated dc bus, ensure proper operation of chargers, and conduct test discharge of batteries.
27.2	125 V dc Station Batteries and Chargers – Load Test	
27.3	250 V dc Station Batteries and Chargers – Load Test	
27.4	Energizing Electrical	Energize to verify proper operation of electrical buses, control centers, and switchgear.
27.5	Energize 500/13 kV Switchyard and 13/4 kV Vital Buses	
27.6	Emergency Power System Operational Test including Diesel Generators and Auxiliaries Operational Tests	To verify the proper operation and independence of redundant vital onsite power systems as suggested in Regulatory Guide 1.41 and proper operation of the diesel generators and associated auxiliary equipment.
28.	Service Water System	To verify the rated head-capacity characteristics of the Service Water System, that the system supplies adequate flow rates through all heat exchangers, and meets the specified requirements when operated in the safeguards mode.
29.	Condenser Air Removal and Priming	To verify flow to all components, and to verify proper operation of instrumentation,

System Tests		Test Objectives
		controllers and alarms. Specifically, each of the systems is tested to ensure: a. All pumps perform their functions satisfactorily. b. All temperature, flow, level and pressure controllers function to control at the required setpoint when supplied with appropriate signals. c. All temperature, flow, level and pressure alarms provide alarms at the required locations when the alarm setpoint is reached and clear when the reset point is reached. d. Design flow rates are established through the principal heat exchangers.
30.	Circulating Water System	Same as 29.
31.	(Not Used)	
32.	Feed System	Same as 29.
32.1	Steam Generator Feed Pump Initial Run	
33.	(Not Used)	
34.	Turbine Auxiliary Cooling	Same as 29.
35.	(Not Used)	
36.	Generator Seal Oil System	Same as 29.
37.	(Not Used)	

	System Tests	Test Objectives
38.	115 Volt AC System	Energize to verify proper operation of 115 V ac buses.
39.	Moisture Separator-Reheater Performance Test	Same as 29.
40.	Turbine Lube Oil System	Same as 29.
41.	Generator Stator Cooling	Same as 29.
42.	Generator Auxiliaries – Isolated Phase Bus (43 through 49 not used)	Same as 29.
50.	Hot Functional Testing Program	Using pump heat, the Reactor Coolant System tested to check heatup and cooldown procedures to demonstrate satisfactory performance of components that are exposed to the reactor coolant temperature; to verify proper operation of instrumentation controllers and alarms, and to provide design operating conditions for checkout of auxiliary systems.
50.1	Reactor Coolant System Heatup	
50.2	Reactor Coolant System at Temperature	
50.3	Reactor Coolant System Cooldown	
50.4	Reactor Coolant System Thermal Expansion	
50.5	Reactor Coolant System Leakage Test – Hot	The Chemical and Volume Control System tested to assure that water can be charged at rated flow against normal Reactor Coolant System pressure; to check letdown flow against design rate for each pressure reduction response of the system to changes in pressurizer level; to check procedures and· components used for boric acid hatching and transfer operations; to check operation of the reactor makeup control; to check operation of the excess letdown and sealwater flowpath; and to verify proper operation and instrumentation, controls, and alarms.
50.6	Pressurizer Power Relief Valve Test	
50.7	Incore Thermocouples and RTD Cross-Calibration	
50.8	Primary Sampling System	
50.9	Pressurizer Pressure and Level Control Test	

System Tests	Test Objectives
50.10 Solid System Pressure Control	
50.11 Steam Generator Safety Valve Test	
50.12 Steam Dump Control	
50.13 Steam Generator Blowdown	The Sampling System was tested to determine that a specified quantity of representative fluid can be obtained safely and at rated conditions from each sampling point.
50.14 Chemical and Volume Control System – Hot	
50.15 Component Cooling System - Hot	The Component Cooling System was tested to evaluate its ability to remove heat from systems and other special equipment; to verify component cooling-cooling flow to all components; and to verify proper operation of instrumentation, controllers, and alarms.
50.16 Initial Turbine Roll - Pump Heat	
50.17 Auxiliary Feed System Performance Test	
50.18 Control Room Inaccessibility	
50.19 Safety Injection Precritical Test	
50.20 Initial Synchronization of Generator	
51. Integrated Test of Engineered Safeguards System and Emergency Power Systems	To demonstrate the composite function of the Engineered Safeguards System by verifying that the proper emergency diesel starts and other related safeguards features.

14.3 Final Station Preparation

The Phase III Plant Operational Test Program and initial fuel loading began when all prerequisite system tests and operations were satisfactorily completed and the facility operating license obtained. Upon completion of fuel loading, the reactor upper internals and pressure vessel head were installed and additional mechanical and electrical operational tests were performed. The purpose of this phase of activities was to prepare the system for nuclear operation and to establish that all design requirements necessary for operation had been achieved. The core loading and post loading tests are described below.

14.3.1 Core Loading

The overall responsibility and direction for initial core loading was exercised by the Station Manager. The overall process of initial core loading was, in general, directed from the operating floor of the Containment Building. Standard procedures for the control of personnel and the maintenance of containment security were established prior to fuel loading. Westinghouse provided technical advisors to assist during the initial core loading operation.

The as-loaded core configuration was specified as part of the core design studies conducted well in advance of station startup and, as such, was not subject to change at startup. In the event that mechanical damage was sustained during core loading operations by a fuel assembly of a type for which no spare was available onsite, an alternate core loading scheme whose characteristics closely approximated those of the initially prescribed pattern would have been determined.

The core was assembled in the reactor vessel, submerged in water containing enough dissolved boric acid to maintain a calculated core effective multiplication constant < 0.95 or a boron concentration of > 2000 ppm, whichever was more restrictive. Core moderator chemistry conditions (particularly, boron concentration) were prescribed in the core loading procedure and were verified periodically by chemical analysis of moderator samples taken prior to and during core loading operation.

Core loading instrumentation consisted of two permanently installed source range (pulse type) nuclear channels and two temporary in-core source range channels plus a third temporary channel which could be used as a spare. The permanent channels when responding were monitored in the Control Room by licensed reactor operators; the temporary channels were installed in the containment structure and were monitored by reactor engineering personnel. At least one permanent channel was equipped with an audible count rate indicator. Both plant channels have the capability of displaying the neutron flux level on strip chart recorders. The temporary channels indicated on rate meters with a minimum of one channel recorded on a strip chart recorder. Minimum count rates of two counts per second, attributable to core neutrons, were required on at least two of the four (i.e., two temporary and two permanent source range detectors) available nuclear source channels at all times following installation of both core sources.

At least two neutron sources were introduced into the core at appropriate specified points in the core loading program to ensure a neutron population of a minimum of two counts per second for adequate monitoring of the core.

Fuel assemblies together with inserted components (control rod assemblies, burnable poison inserts, source spider, or thimble plugging devices) were placed in the reactor vessel one at a time according to a previously established and approved sequence which was developed to provide reliable core monitoring with minimum possibility of core mechanical damage. The core loading procedure documents include a detailed tabular check sheet which prescribed and verified the successive movements of each fuel assembly and its specified inserts from its initial position in the storage racks to its final position in the core. Checks were made of component serial numbers and types at successive transfer points to guard against possible inadvertent exchanges or substitutions of components, and at least two fuel assembly status boards were maintained throughout the core loading operation.

An initial nucleus of eight fuel assemblies, the first of which contained a neutron source, was the minimum source-fuel nucleus which permitted subsequent meaningful inverse count rate monitoring. This initial nucleus was determined by calculation and previous experience to be markedly subcritical under the required conditions of loading.

Each subsequent fuel addition was accompanied by detailed neutron count rate monitoring to determine that the just loaded fuel assembly did not excessively increase the count rate and that the extrapolated inverse count rate ratio was not decreasing for unexplained reasons. The results of each loading step were evaluated by Public Service Electric and Gas (PSE&G) and its technical advisors before the next prescribed step was started.

Criteria for safe loading required that loading operations stop immediately if:

1. An unanticipated increase in the neutron count rates by a factor of two occurred during any single loading step after the initial nucleus of eight fuel assemblies were loaded (excluding anticipated change due to detector and/or source movement).

2. The neutron count rate on any individual nuclear channel increased by a factor of five during any single loading step after the initial nucleus of eight fuel assemblies were loaded (excluding anticipated changes due to detector and/or source movements).

An alarm in the containment and Control Room was coupled to the source range channels with a setpoint at five times the current count rate.

This alarm would automatically alert the loading operation to an indication of high count rate and required an immediate stop of all operations until the situation had been evaluated. The alarm used for this purpose was the containment evacuation alarm. In the event the evacuation alarm was actuated during core loading and after it had been determined that no hazards to personnel existed, special preselected personnel were permitted to remain in the containment vessel to evaluate the cause and determine future action. The preselected who were allowed to remain in the containment vessels were the Senior Reactor Operator designated by PSE&G and its technical advisers.

Core loading procedures specified alignment of fluid systems to prevent inadvertent dilution of the reactor coolant, restrict the movement of fuel to preclude the possibility of mechanical damage, prescribe the conditions under which loading could proceed and provide for continuous fuel and fuel assembly insert identification.

14.3.2 Post Loading Tests

Upon completion of core loading, the reactor upper internals and the pressure vessel head were installed and additional operational, mechanical, and electrical tests were performed prior to initial criticality. The final pressure tests were conducted after filling and venting was completed.

Mechanical and electrical tests were performed on the control rod drive mechanisms (CRDMS). These tests included a complete operational checkout of the mechanisms. Checks were made to ensure that the control rod assembly position indicator coil stacks were connected to their position indicators. Similar checks were made on CRDM coils.

Tests were performed on the reactor trip circuits to test manual trip operation, and actual control rod assembly drop times were measured for each control rod assembly. By use of dummy signals, the Reactor Control and Protection System was made to produce trip signals for the various unit abnormalities that require tripping.

At all times that the CRDMS were being tested, the boron concentration in the coolant-moderator was high enough that criticality could not be achieved with all control rod assemblies out.

A complete functional electrical and mechanical check was made of the in-core nuclear flux mapping system at operating temperature and pressure.

A listing of tests required prior to initial criticality is contained in Table 14.3-1.

	System Tests	Test Objectives
1.	Reactor Coolant System Leakage Test	To verify proper sealing of reactor vessel closure head after fuel load.
2.	RTD Bypass Loop Flow Verification	To measure flow and verify acceptable transport times.
3.	Reactor Coolant Flow Coastdown	To measure flow rate changes and associated delay times subsequent to various pump stops.
4.	Pressurizer Spray and Heater Capability and Continuous Spray Flow Setting	To verify pressurizer effectiveness and establish continuous spray flow rate.
5.	Reactor Coolant System Flow Measurement	To interrelate pump input power and loop P as measurement of actual RCS flow rate.
6.	Rod Position Indication System	Prior to core loading, test signals were used to check the system response and verify correct indicating and control functions. After fuel loading and after the position indication coils were installed, a calibration and complete operational check was performed by operating individual control rod drive mechanisms.
7.	Rod Drive Mechanism	To verify the proper timing of each Rod Control System slave cycler and conduct an operational check of each full length mechanism with RCCA attached.
8.	Rod Drop Time Measurement	To determine drop time of full length rods under full flow, no flow, cold and hot conditions.
9.	Rod Control System Operation	To demonstrate that system satisfactorily performs the required control and indication functions to verify ready for initial criticality.
10.	Part Length Rod Mechanism Brake Test	To verify proper operation of brake arms in each part length mechanism.
11.	In-Core Flux Mapping System	To verify proper operation of the in-core movable detectors.

14.4 Initial Testing of the Operating Reactor

After satisfactory completion of fuel loading and final precritical tests, nuclear operation of the reactor began. The Phase III Plant Operational Testing Program continued through initial criticality, low power testing, and power level escalation. The purpose of these tests was to establish the operational characteristics of the reactor core, to acquire data for the proper calibration of setpoints, and to ensure that operation will be within license requirements. A brief description of the testing is presented in the following sections. Table 14.4-1 summarizes the tests which were performed from initial criticality to rated power.

In the development of the Phase III testing program, PSE&G complies with the intent of Regulatory Guide 1.68, dated November 1973, with the following exceptions:

1. Natural Circulation Test – This test is performed on a first-of-a-kind plant only, and therefore will not be performed on Salem Unit 1. Test results were submitted to the NRC for the Trojan 1 reactor plant, which satisfactorily performed a natural circulation test.

   PSE&G satisfactorily performed Natural Circulation testing on Salem Unit 2 between August 23 and August 29, 1980, as part of a Special Low Power Test Program required by conditions 2.C(6)b and c of Operating License DPR-75. The Test Program consisted of the following tests:

SUP 90.1	Natural Circulation Test
SUP 90.2	Natural Circulation with Simulated Loss of Offsite AC Power
SUP 90.3	Natural Circulation with Loss of Pressurizer Heaters
SUP 90.4	Effect of Steam Generator Secondary Side Isolation on Natural Circulation
SUP 90.5	Natural Circulation at Reduced Pressure
SUP 90.6	Cooldown Capability of Charging and Letdown
SUP 90.7	Simulated Loss of All Onsite and Offsite AC Power

   A test Summary was transmitted to the NRC in a Letter dated September 8, 1980.

2. Loss-of-Flow Test – The reactor coolant flow coastdown test performed during Phase III testing will provide data on loss-of-flow characteristics. The plant trip test (see Table 14.4-1) performed during Phase III testing will provide additional Reactor Coolant System response data. Sufficient information will be made available from these tests so that a special loss-of-flow test at power will be unnecessary.

3. Vibration Measurements on Reactor Internals – Extensive vibration testing and a detailed vibration analysis are performed on each first-of-a-kind unit only and reported to the NRC. Indian Point Unit 2 has been established as the prototype for all four-loop plants. An extensive inspection of reactor internals will be performed in lieu of vibration measurements.

4. Pressure Coefficient of Reactivity Measurements – Direct measurements of the pressure coefficient of reactivity are not necessary, since the change in reactivity (PCM) over the entire pressure range is so small that such a test is of no practical value.

5. Axial Xenon Suppression Test (Part-Length Rod Effectiveness) – Determination of part-length effectiveness for controlling xenon transients is not necessary since use of part-length rods is prohibited and they will not be installed. Constant axial offset control required by the Technical Specifications is the method of controlling xenon transients.

6. Turbine Trip – This test will be performed in conjunction with the generator trip test, which will be performed at 100 percent of rated thermal power. It is expected that a reactor trip and a turbine trip will result within a reasonable period of time following opening of the generator main breaker. During performance of this test, the time delay between the generator trip and the turbine trip, as well as any turbine overspeed, will be noted and recorded.

6a. Process Computer – This test will not be performed since the process computer performs no control or protection function. It serves only as a data logger.

7. Dynamic Rod Drop Test – The purpose of including this test in previous test programs has been to test the response of the automatic turbine runback feature. This control feature is not incorporated into the Salem design. Plant response to a dropped-rod casualty will therefore be tested by performing a rod drop test, which also will test the negative rate trip circuitry. The test will be performed as part of the at-power testing sequence (see Table 14.4-1).

Note: Salem NRC License Amendment 278-261 (Salem 1 and 2, respectively) approved the removal of the Negative Flux Rate Trip. This function was initially disabled by setting the setpoint to a greater value than the Maximum Negative Rate expected (per design change package (DCP) 80094424). The Negative Flux Rate Trip circuitry has been physically removed from both Unit 1 and 2 per DCPs 80097106 and 80099680.

14.4.1 Initial Criticality

Initial criticality was established by sequentially withdrawing the shutdown and control groups of control rod assemblies from the core, leaving the last withdrawn control group inserted far enough in the core to provide effective control when criticality was achieved, and then continuously diluting the heavily borated reactor coolant until the chain reaction was self-sustaining. Successive stages of control rod assembly group withdrawal and of boron concentration reduction were monitored by observing changes in neutron count rate as indicated by the regular source range nuclear instrumentation as functions of group position during rod motion and, subsequently, a reactor coolant boron concentration and primary water addition to the Reactor Coolant System during dilution. Throughout this period periodic samples of the primary coolant boron concentration were obtained and analyzed.

Primary safety reliance was based on inverse count rate ratio monitoring as an indication of the nearness and rate of approach to criticality of the core during control rod assembly group withdrawal and during reactor coolant boron dilution. The rate of approach was reduced as the reactor approached extrapolated criticality to ensure that effective control was maintained at all times.

Written procedures specified alignment of fluid systems to allow controlled start and stop and adjustment of the rate at which the approach to criticality was expected, and identify chains of responsibility and authority during reactor operations.

14.4.2 Lower Power Testing

A prescribed program of reactor physics measurements was undertaken to verify that the basic static and kinetic characteristics of the core were as expected and that the values of the kinetic coefficients assumed in the safeguards analysis were indeed conservative.

The measurements were made at lower power and primarily at or near operating temperature and pressure. Measurements which were made included verification of calculated values of control rod assembly group reactivity worths, is thermal temperature coefficient under various core conditions, differential boron concentration reactivity worth and critical boron concentrations as functions of control rod assembly group configuration. In addition, relative power distribution measurements were made. Concurrent tests were conducted on the instrumentation including the source and intermediate range nuclear channels.

Detailed procedures were prepared to specify the sequence of tests and measurements to be conducted and the conditions under which each was to be performed to ensure both safety of operation and the relevancy and consistency of the results obtained.

14.4.3 Power Level Escalation

When the core performance characteristics of the reactor were verified by the low power testing, a program of power level escalation in successive stages brought the unit to its full rated power level. Both reactor and unit operational characteristics were closely examined at each stage and the relevance of the safeguards analysis verified before escalation to the next program level was effected.

Measurements were made to determine the relative power distribution in the core as functions of power level and control assembly group position. These measurements supplied additional core performance data in terms of heat flux and margins to departure from nucleate boiling.

Secondary system heat balances ensured that the several indications of power level were consistent and provided bases for calibration of the power range nuclear channels. The ability of the Reactor Control System to respond effectively to signals from primary and secondary instrumentation under a variety of conditions encountered in normal operation was verified.

The dynamic response characteristics of the reactor coolant and steam systems were evaluated at prescribed power levels. The responses of system components were measured for 10-percent reduction of load and recovery, 50-percent reduction of load, plant trip, and trip of two control rods.

Adequacy of radiation shielding was verified by gamma and neutron radiation surveys inside the containment and throughout the station site.

The sequence of tests, measurements, and intervening operations was prescribed in the power escalation procedures together with specific details relating to the conduct of the several tests and measurements.

14.4.4 Post Startup Surveillance and Testing Requirements

Post startup surveillance and testing requirements were designed to provide assurance that essential systems, which include equipment components and instrument channels, are always capable of functioning in accordance with their original design criteria. These requirements can be separated into two categories:

1. The system must be capable of performing its function, i.e., pumps deliver at design flow and head, and instrument channels respond to initiating signals within design calibration and time response.

2. Reliability is maintained at levels comparable to those established in the design criteria and during early station life.

The testing requirements, as described in the Technical Specifications, establish this reliability and, in addition, provide the means by which this reliability is continually confirmed. Verification of operation of complete systems is checked at refueling intervals. Individual checks of components and instrumentation are made at more frequent intervals as outlined in the Technical Specifications.

The techniques used for testing the instrument channels included a preoperational calibration which confirmed values obtained during factory test programs. These reconfirmed calibration values became the reference for recalibration maintenance at refueling intervals during station life. Periodic testing, as defined to the Technical Specifications, includes the insertion of a predetermined signal that trips the channel bistable. Indication of the operation is confirmed and recorded.

Testing of components is initiated through manual actuation. If response times are of importance, they are measured and recorded. The capability to deliver design output is checked with instrumentation and compared against design data. Allowable deviations have been established in the Technical Specifications. The component is operated a sufficient length of time to allow equalization of operating temperatures in bearings, seals, and motors. These parameters are checked periodically. The component is surveyed for excessive vibration and readings are recorded.

Public Service Electric & Gas believes that testing in accordance with the program described above provides a realistic basis for determining maintenance requirements and, as such, ensures continued system capabilities, including reliability, equal to those established in the original criteria.

14.4.5 Safety Precautions

The test operations during low power and power escalation were similar to normal station operation at power, and normal safety precautions were observed. Those tests which required special operating conditions were accomplished using test procedures which prescribed necessary limitations and precautions.

Test	Conditions	Objectives	Comments
Nuclear Design Check Tests (1)	Hot zero power	To verify that nuclear design predictions for endpoint boron concentration, temperature coefficient and normal flux distribution are valid.	SAR criteria applicable
Rod and Boron Worth Measurements during Boron Dilution and Addition (1)	Hot zero power	a) To determine differential and integral worth of control banks b) To determine differential boron worth over range of control bank motion.	SAR criteria applicable
RCCA Pseudo Ejection at Zero Power (1)	Hot zero power	To determine worth of most reactive RCCA in the rod configuration assumed in the accident analysis.	The "just critical" boron concentration and flux distribution also obtained with the most reactive RCCA withdrawn
Minimum Shutdown Verification and Stuck Rod Worth Measurements (1)	Hot zero power	To measure the minimum shutdown boron concentration with one stuck control rod assembly and measure integral worth of all rod banks.	Verify stuck control rod assembly shutdown criteria
Calibration of Steam and Feedwater Flow Instrumentation at Power	Hot zero power and 30%, 50%, 75%, and 100%	To calibrate feedwater and steam flow instruments for proper indication.	Verify correct inputs to control systems
Natural Circulation (Unit 2 only)	<7%	To demonstrate natural circulation heat removal capability and provide plant response information, baseline data for specific plant characteristics, and supplemental operator training.	Testing successfully completed between 8/23 and 8/29/80
Chemistry and Radiochemistry Tests	Specific analyses at low power and 30%, 50%, 75%, and 100% power	To demonstrate ability to control water quality.	Ability to control water quality verified at each power level

Test	Conditions	Objectives	Comments
Radiation Monitoring and Shielding Evaluation	Low power and during power escalation at 30%, 70%, 100%	To measure and record radiation levels in accessible areas of the nuclear plant.	Radiation levels were verified to be within the shielding design criteria
Effluent Monitoring Systems	As early in power operation as possible and repeated after operation at 30%, 50%, 75%, and 100% power	To verify calibration of effluent monitors by lab analysis of radioactive waste samples.	Verification of calibration of plant effluent monitors
Power Coefficient and Integral Power Defect Measurement (1)	Various power levels up to 100% power	To verify validity of nuclear design predictions for differential power coefficient and determine the integral power defect.	SAR criteria applicable
Dynamic Automatic Steam Dump Control	<10% power	To demonstrate proper operation of steam dump control and verify setpoints.	Steam dump operation satisfies design criteria
Automatic Steam Generator Level Control	30% power	To demonstrate satisfactory performance of the automatic steam generator level control system.	System maintained steam generator level during simulated transient
Turbine Overspeed Trip	Approximately 10% power	To verify that turbine overspeed trip setpoint is correct.	
Turbine Control System Checkout and Startup Adjustments of Reactor Control System	0%, 30%, 50%, 75%, 100%	Defines turbine steamline inlet pressure characteristic curve. Reprograms rod control system Tavg program for designed stm press.	Calibrated control systems in accordance with design specifications
Automatic Reactor Control temperature within design requirements	30% power	To verify reactor control system performance characteristics.	System maintained reactor average

Test	Conditions	Objectives	Comments
Steam Generator Moisture Carryover	75%, 90%, and 100% power	To determine average total moisture carryover from steam generators during steady state operation.	Calculated steam moisture carryover had to be within design limits
Incore/Excore Detector Calibration(1)	75% power	To determine power distribution and excore response at various rod configurations.	Conformance with design requirements verified
Nuclear and Temperature Instrumentation Calibration and Thermal Power Measurement(1)	30%, 50%, 75%, 90%, and 100% power	a) To determine linearity of power range detectors. b) To calibrate power range channels to reflect thermal power level. c) To obtain nuclear instrument overlap data. d) To obtain temperature data for aligning and setting temperature and T instruments and setpoints.	Calibrate instruments in accordance with design specifications
Static RCCA Drop and RCCA Below Bank Position Measurements(1)	50% power	a) To verify dropped rod worth and hot channel factors assumed in SAR and evaluation of effects on DNB. b) To demonstrate detectability of a single RCCA below bank position or dropped.	SAR criteria applicable
RCCA Pseudo Ejection and RCCA above Bank Position Measurements(1)	50% power	a) To verify ejected-rod worth and hot channel factors assumed in SAR. b) To demonstrate detectability of a single RCCA above bank position or ejected.	Plant parameter variations are within design limits
Large Load Reduction Tests	Large step (50% reduction of load at 75%, 100% power)	To demonstrate plant response to design large-load changes.	Plant parameter variations are within acceptance limits
Shutdown from outside Control Room	10% power	After plant trip from outside control room, verify ability to maintain hot shutdown conditions from outside control room.	Plant parameters can be controlled within design limits from outside control room
Loss of Offsite Power	10% power	To demonstrate plant response upon occurrence of a loss of offsite power.	Plant parameter variations are within design limits

Test	Conditions	Objectives	Comments
Generator Trip	100% power	To determine any turbine overspeed and to demonstrate plant response to a trip from 100% power.	Plant parameter variations are within design limits
Rod Drop and Plant Trip (2)	50% power	To demonstrate plant response and control system behavior to a two-rod drop and subsequent plant trip; to demonstrate the functioning of the negative rate trip circuitry.	Plant parameter variations are within design limits
NSSS Acceptance Test	100% power	To demonstrate satisfactory operation of the NSSS during a 100 hour full power run.	Plant parameter variations are within design limits

Note:

(1) These tests verified that core performance is within design predictions.
(2) Salem NRC License Amendment 278-261 (Salem 1 and 2 respectively) approved the removal of the Negative Flux Rate Trip. This function was initially disabled by setting the setpoint to a greater value than the Maximum Negative Rate expected (per design change package (DCP) 80094424). The Negative Flux Rate Trip circuitry has been physically removed from both Unit 1 and 2 per DCPs 80097106 and 80099680.

14.5 Test Program Organization – Unit 1

14.5.1 Organization and Responsibility

The Electric Production Department had responsibility for the Startup and Test Program through the Station Manager who, in turn, was directed by the Project Manager for startup-related activities. This responsibility was discharged through the Salem Startup Group (SSG) who established the detail, specific requirements for schedule and testing in support of project schedule requirements; and coordinated and directed, through established organizational structures, all contributing parties responsible for specific activities within the Startup and Test Program.

The SSG was assisted, as required, through liaison with other Company departments, such as Engineering & Construction, Quality Assurance and Energy Laboratory, as shown on Figure 14.5-1.

United Engineers & Constructors (UE&C), under the direction of Public Service Electric & Gas (PSE&G), coordinated construction schedules with testing requirements, provided manpower support as required, corrected deficiencies or made repairs, and wrote and performed certain Phase I test procedures as directed by the SSG Test Engineers.

Westinghouse provided onsite technical consultation to assure that the safety and reliability of Westinghouse supplied systems was not compromised and ensured that equipment was adequately tested. Westinghouse had neither the responsibility for supervision of PSE&G or UE&C personnel nor direct responsibility for planning, scheduling, or management of testing activities. All testing activities on safety-related equipment was independently audited by QA/QC groups to ensure that any deviations from Final Safety Analysis Report commitments, design requirements or established operating and administrative procedures were identified, evaluated, and documented. Due to the nature of Phase III testing, the station operating staff directed Phase III tests. Test procedure preparation, coordination, and scheduling of Phase III activities remained under the control of the SSG.

The SSG coordinated and directed, through established organizational structures, all contributing parties responsible for specific activities within the Preoperational and Startup Testing Program.

The SSG was comprised of test execution, test administration, and construction turnover support groups. The Startup Group Head directed the SSG. Reporting to the Startup Group Head were two Lead Startup Engineers (one for Phase I/II test execution and one for test administration) and the Construction Turnover Superintendent.

The Construction Turnover Group was comprised of supervisors and coordinators assigned to coordinating crafts in supporting certain construction-related tests and accomplishing cleanup construction activities.

The test execution and administration groups are headed by Startup Engineers responsible for the supervision, coordination, detailed guidance, and direction for assigned Test Engineers. The SSG Test Engineers were charged with the following general responsibilities as appropriate:

1. Follow and expedite construction progress to support startup requirements
2. expedite testing progress regarding the requirements of Project Planning and Scheduling documents
3. Define testing boundaries and requirements
4. Write and/or ensure that detailed preoperational and startup test procedures are available to all contributing parties
5. Schedule daily testing activities and technically direct their execution
6. Ensure proper documentation of test results

To accomplish these tasks, Test Engineers were onsite full time and had no other duties outside the testing program.

For testing activities commencing with core load, the Station Chief Engineer managed and directed all activities. Members of the SSG and Test Engineer groups participated in Phase III operations, assisting the Electric Production Department as required.

Typical organization chart for the SSG is presented in Figure 14.5-2.

The following specific responsibilities apply to the Startup Engineers and Test Engineers:

<u>Startup Engineers</u>

1. Review and provide startup input to the Master Project CPM and coordinate the development of weekly/daily testing schedules by the Test Engineers.
2. Primary responsibility for establishing test activity target dates and priorities. Assist in identifying required construction progress necessary to support these dates.
3. Analyze, in detail, and develop specific administrative/procedural methods and manpower recommendations to satisfactorily accomplish the startup program.
4. Control and periodically revise the Salem Startup Manual thereby keeping overall startup policies, practices, procedures, and scope of testing up-to-date.
5. Coordinate and direct assigned personnel and all contributing testing parties in the performance of their assigned responsibilities.

6. Conduct test planning meetings to inform all contributing parties of long-term requirements, identify startup activities requiring schedule or quality improvement, and resolve, or develop a course of resolution for major startup-related problems.

7. Onsite authority regarding resolution of detailed testing prerequisites, schedule, scope, methods, or technical conflicts rests with the SSG.

8. Establish by means of document control, a smooth, complete, and efficient flow of startup-related information, procedures, test results, turnover and acceptance reports.

9. Review startup-related directives, instructions, and general or detailed procedures written by all contributing parties to assure compatibility and consistency with Salem Startup Manual. Periodically audit practices required by these documents for conformance.

10. Release all Phase II Startup Procedures for test. Certify completion of same.

11. Schedule and coordinate the site activities of Westinghouse specialists and other vendor representatives.

12. Review and approve detailed Phase I test procedures generated by Test Engineers.

13. Maintain close coordination with management representatives of all contributing parties.

Test Engineers

1. Through early advanced planning, develop detailed test activity schedules, test manpower requirements, and prerequisites. Coordinate adherence to same.

2. Determine the nature and degree of testing required in accordance with Engineering specifications and standard procedures.

3. Develop, assemble, initiate Test Requirement Outlines, Testing Boundary Diagrams, Preoperational Testing Notifications, Operational Testing Turnovers, and other project control documents, for testing activities on assigned systems.

4. Schedule and expedite procedure/documentation generated by others and construction completion dates for startup activities.

5. Identify and specify special (temporary) equipment necessary to support testing.

6. Administer testing identification tagging and temporary equipment usage on assigned systems.

7. Write cleaning procedures and other preoperational test procedures in accordance with Engineering and Construction Department (E&CD) technical requirements. Review and accept/or approve Phase I detail procedures written by others.

8. Formally notify all contributing parties of required test, turnover, and acceptance dates in a timely manner.

9. Schedule vendor service representatives in support of test activities, as required.

10. Direct all contributing parties through existing organizational structures during the preparation for and execution of testing activities on assigned systems.

11. Maintain marked "Information Only" diagrams showing test activity completion status.

12. Review test results for assigned systems.

13. Write assigned Phase II and III Startup Procedures. Review other procedures which interface with assigned system.

14. Maintain the official Master Startup Procedure during the execution of Phase II testing.

15. Prepare test reports, review and coordinate evaluation of test results.

16. Maintain the official documentation files.

14.5.2 Test Procedure Preparation and Review

All tests were performed in strict accordance with approved written test procedures. These test procedures included provisions to ensure that prerequisites were met, adequate protective instrumentation was used, and required test monitoring and documentation were performed.

Phase I test procedures were written under the direction of the SSG by various contributing groups within PSE&G and UE&C. Procedures were reviewed by the originator, SSG, representatives of the E&CD, and the Quality Assurance Department, as appropriate.

Phase II and III startup procedures were written by individuals designated by the SSG. Final specific approval of the test procedures were given by the Station Manager. Approval of the procedure was given after formal review by the following personnel:

1. Author

2. Westinghouse Representative (for NSSS tests)

3. Engineering Department Representatives (for Phase II tests)

4. Quality Assurance Representative (for safety-related tests)

5. Station Operations Review Committee (SORC)

The approved startup procedure was retained by the SSG until the scheduled test date. At this time the prerequisites were verified and the procedure, or sections of it, were released for test by the Lead Startup Engineer. Phase III startup procedures were released by the Reactor Engineer.

The SORC completed its review of all scheduled safety-related system and core loading tests prior to loading fuel and of all post-core loading tests prior to initial criticality. Furthermore, the SORC advised the station Superintendent in writing when required changes and additions to the Plant Manual resulting from these tests had been properly entered.

All completed test procedures and accompanying data were made a part of the station's historical record and, therefore, are retained in the form of Test Reports.

A specific format was prescribed for the preparation of written startup test procedures. The procedures included a statement of test objectives, a list of references used in preparation, and a list of prerequisites and initial conditions to be established for each test.

Required test or special equipment was specified, and unusual environmental conditions required or generated by the test were described. The specific acceptance criteria for determining the success or failure of the test was clearly identified (where appropriate) and was a part of the test procedure. Any general precautions or limitations imposed by operational or safety requirements were also specified.

A detailed step-by-step procedure was provided for each Phase II and III test. Test Engineer's initials were used to document completion of each step. Where special conditions such as abnormal valving or use of jumpers or bypasses are necessary, control measures are specified to insure that the abnormal configuration is returned to normal upon completion of the test.

The procedures included a section for recommended changes to plant operating procedures and the Plant Manual that were generated as a result of the tests. The procedures also contained a section for remarks related to the test. A standard cover sheet was developed to facilitate formal test procedure review and approval.

14.5.3 Startup Procedure Changes

Changes to approved startup test procedures were reviewed and approved by the same functional groups involved in the original document. A change notice was issued which included the change(s) and reason(s) for the change(s).

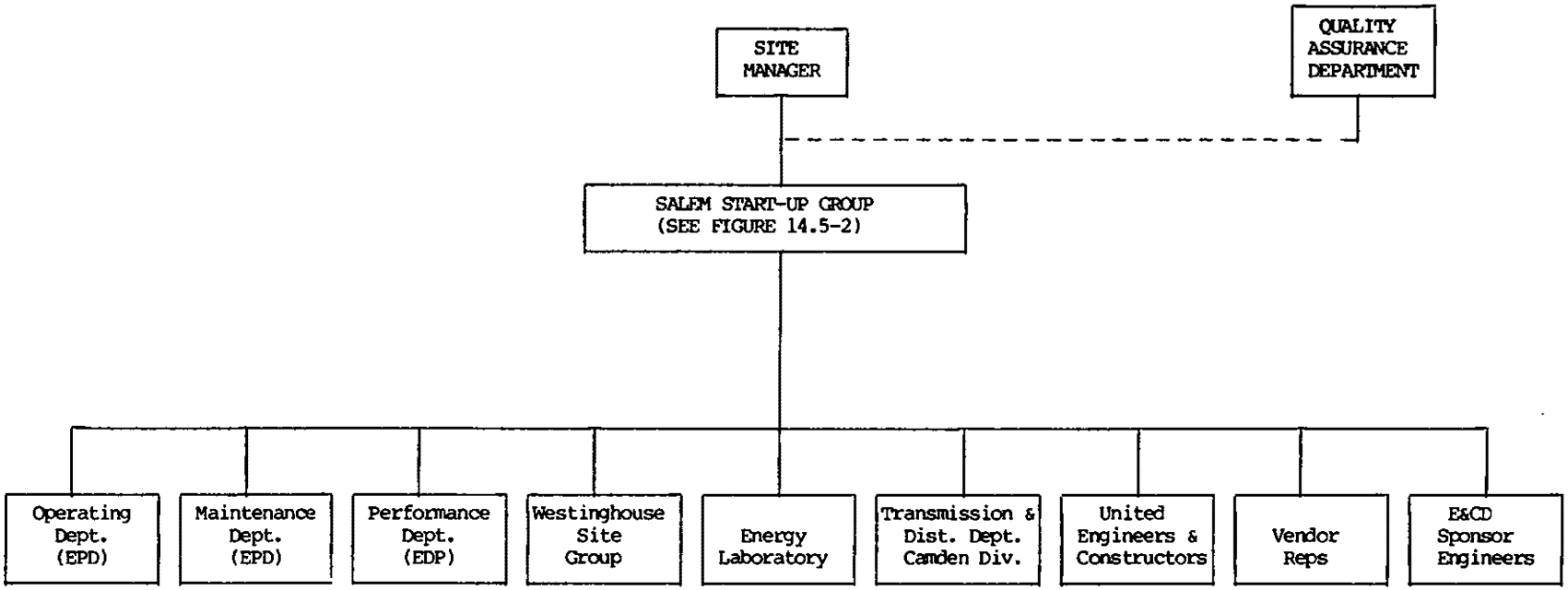
On-the-spot changes to startup procedures were authorized by the Station Manager, or his designated representative, only if the change did not modify the original intent of the procedure. However, a post-change review was subsequently required to be conducted by the same functional groups involved in the original document.

14.5.4 Startup Test Results

Test results were reviewed by the same functional groups who reviewed and approved the original test procedure. The Lead Startup Engineer initiated the review process by ensuring that the procedure had been completed and that any exceptions or comments listed by the Test Engineer who conducted the test were clearly explained. He then forwarded the results to the following personnel (in the order listed):

1. Reactor Engineer (for physics and fuel handling tests)
2. Westinghouse Representative (for NSSS tests)
3. Engineering Department Representatives (for Phase II tests)
4. Quality Assurance Representative (for safety-related tests)
5. SORC Chairman
6. Station Manager (for final approval)

Copies of the final approved test procedure and results were forwarded to the appropriate Quality organization for record retention.



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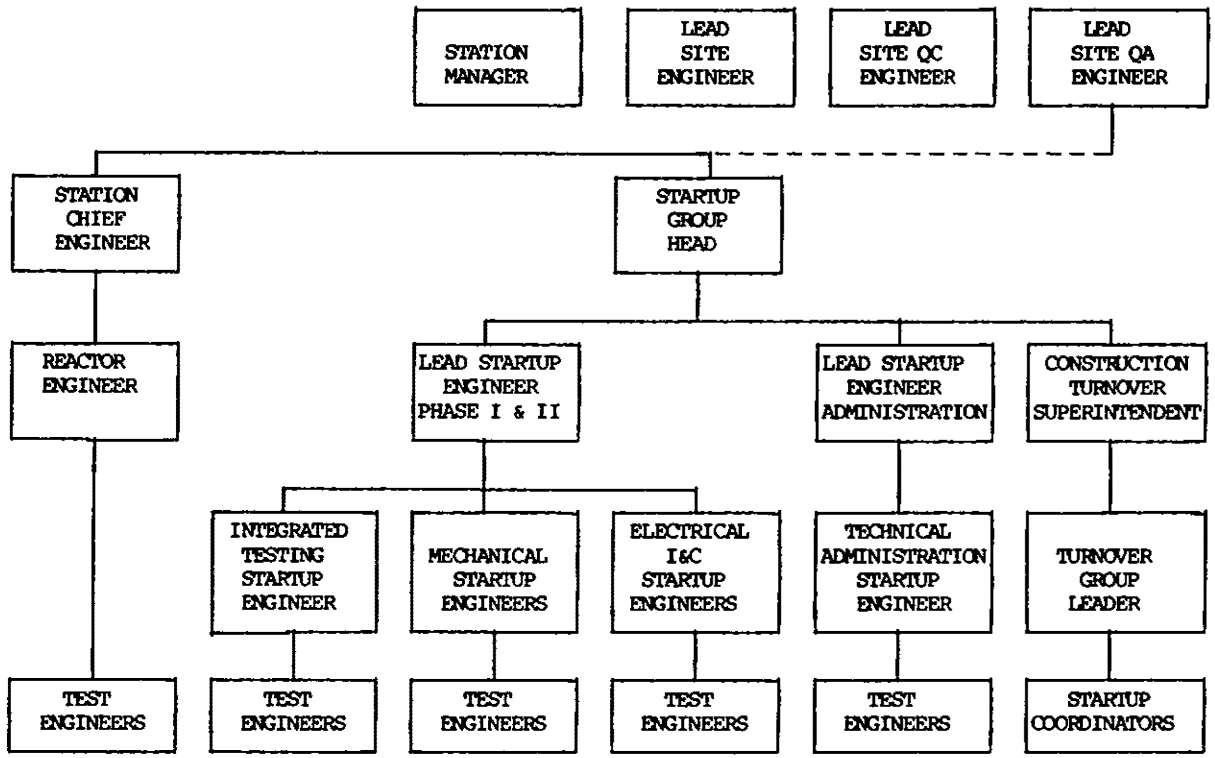
PUBLIC SERVICE ELECTRIC AND GAS COMPANY SALEM NUCLEAR GENERATING STATION	Startup Program Organization - No. 1 Unit
	Updated FSAR — Figure 14.5-1



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PUBLIC SERVICE ELECTRIC AND GAS COMPANY SALEM NUCLEAR GENERATING STATION	Typical Startup Group Organization - No. 1 Unit
	Updated FSAR — Figure 14.5-2

14.6 Test Program Organization – Unit 2

14.6.1 Organization and Responsibility

The Startup and Test Program encompassed all startup and testing activities for components, systems and facilities from completion of construction activities to final acceptance of the entire unit. The program was divided into three phases, defined as follows:

Phase I Construction Verification Tests

Phase II Preoperational Tests

Phase III Plant Operational Tests

For Unit 2, the titles of the Phase I and II test phases were changed to reflect more clearly industry standards and the division of responsibility at Salem. The scope of testing, however, remained essentially the same as defined in Section 14.1.

The overall responsibility for the startup and test program rested with Public Service Electric & Gas (PSE&G). Specific responsibility for Phases I and II rested with the Engineering and Construction Department (E&CD) through the Project Construction Manager. Specific responsibility for Phase III rested with the Electric Production Department (EPD) through the Manager – Salem Generating Station. The responsibility for all phases was discharged through the Salem Startup Group (SSG), which established the detail, specific requirements for scheduling and testing in support of project requirements; and coordinated and directed, through established organizational structures, all contributing parties responsible for specific activities within the Startup and Test Program. The SSG was composed of two subgroups (see Figure 14.6-2).

The United Engineers & Constructors (UE&C) Test Group (UTG) had the primary responsibility for the Phase I Program. The PSE&G Startup Group (PSSUG) had primary responsibility for the Phase II and III Programs.

The UTG was directed by a Lead Test Engineer and consisted of a Mechanical Group, an Instrument and Controls Group and an Electrical Group, each having a supervisor and staffed with test engineers as required.

The PSSUG was directed by the Startup Engineer and consisted to an Instrumentation Controls and Electrical (ICE) Group, a Mechanical and Integrated (MI) Test Group, and a Technical Administration Group. Each group was directed by a Senior Construction Engineer and staffed as required with test engineers. The PSSUG Startup Engineer also provided technical direction to the UTG.

The SSG was assisted, as required, through liaison with other Company departments, such as Engineering, Construction, Quality Assurance, and the Energy Laboratory, as shown on Figure 14.6-1.

Westinghouse provided onsite technical consultation to assure that the safety and reliability of Westinghouse supplied systems were not compromised and that equipment was adequately tested.

Westinghouse had neither the responsibility for supervision of PSE&G or UE&C personnel nor direct responsibility for planning, scheduling, or management of testing activities.

The objective of the Phase I Construction Verification Test Program was to: 1) verify that fabrication, construction and erection of plant systems and equipment had been performed in accordance with applicable codes, standards, drawings, specifications, procedures and good engineering practice; 2) to verify individual component cleanliness, integrity and intended functional capability; and 3) to bring together completed systems, functionally checked and ready to support preoperational testing.

This program included pressure testing, cleaning, flushing, and initial running of equipment to support flushing, initial energization of electrical equipment, loop checks and instrument calibration. There were certain exceptions, however, to the UTG's scope of Phase I testing. The E&CD retained primary responsibility for planning and implementation of the Phase I Construction Verification Test Program for the following systems:

1. Computer System
2. Reactor Protection System
3. Process Control Systems
4. Radiation Monitoring System
5. Turbine Supervisory Instrumentation System
6. Loose Parts Monitoring System
7. Status Panel
8. Security System
9. Safeguards Equipment Control System
10. Nuclear Instrumentation System
11. Annunciator System
12. DC Power Systems
13. Rod Control System
14. AC Power Distribution Systems

15. 115 V AC System

16. Turbine Hydraulic Control System

The Phase II Preoperational Test Program consisted of individual and multi-system tests through hot functional tests up to initial core load. They were performed to ensure system conformance to engineering design parameters. Primary responsibility for this program rested with the E&CD.

The Phase III Plant Operational Test Program consisted of integrated system tests from the time of initial core load through plant acceptance tests. They were performed to establish unit operational characteristics and ensure conformance with licensing requirements. Primary responsibility for this program rested with the EPD.

The Quality Assurance Department verified, through their participation in the Startup Program, that applicable regulatory requirements were observed in the preparation and review of procedures, conduct of inspections and tests, recording of data, disposition of nonconformances, performances of repairs, maintenance, storage and preservation of equipment and maintenance of records.

14.6.2 Test Procedure Preparation and Review

All tests were performed in strict accordance with approved written test procedures. These test procedures included provisions to ensure that prerequisites were met, adequate protective instrumentation was used and required test monitoring and documentation were performed.

Phase I test procedures within the scope of UE&C were written under the direction of UE&C Lead Test Engineer. Procedures were reviewed and approved by the originator, the UE&C Lead Test Engineer, and the PSSUG Startup Engineer. Phase I test procedures within the scope of PSE&G were written under the direction of the PSSUG Startup Engineer. Site Quality Assurance approved all standard procedures and all detail procedures for safety-related systems which were not based upon standard procedures.

Phase II startup procedures were written under the direction of the PSSUG Startup Engineer. Final approval to commence testing was given by him after formal review by the following personnel:

1. Author

2. Westinghouse Representative (for NSSS tests)

3. Quality Assurance Representative (for safety-related tests)

4. Preoperational Testing Review Committee (PORC)

The PORC consisted of personnel from the Engineering Department, Production Department, and the PSSUG. The Committee was chaired by a representative of the Engineering Department.

Phase III startup procedures were the responsibility of the EPD. The procedures were written by either EPD or by the PSSUG under the direction of the PSSUG startup engineer. Final approval to commence testing was given by the Station Manager after formal review by the following personnel:

1. Author
2. Westinghouse Representative (for NSSS tests)
3. Quality Assurance Representative (for safety-related tests)
4. Station Operations Review Committee (SORC)

The SORC completed its review of all tests to be performed prior to initial criticality (Table 14.3-1) prior to loading fuel and completed its review of all post criticality tests (Table 14.4-1) prior to initial criticality. Furthermore, the SORC advised the Station Manager in writing when changes or additions to the station plant manual that resulted from the tests had been properly entered.

All completed test procedures and accompanying data were retained in the form of test reports and are a part of the station's historical record.

The approved startup procedures were retained by the PSSUG until the scheduled test date. At this time, the prerequisites were verified and the procedures were released for test by the PSSUG Startup Engineer.

A specific format was prescribed for the preparation of startup test procedures. The procedures included a statement of test objectives, a list of references used in preparation and a list of prerequisites and initial conditions to be established for each test. Required test or special equipment was specified, and unusual environmental conditions required or generated by the test were described. The specific acceptance criteria for determining the success or failure of the tests were clearly identified (where appropriate) and became a part of the test procedure. Any general precautions or limitations imposed by operational or safety requirements were also specified.

A detailed step-by-step procedure was provided for each Phase II and III test. The test engineer's initials were used to document completion of each step. Where special conditions, such as abnormal valving, or use of jumpers or bypasses became necessary, control measures were specified to ensure that the abnormal configuration was returned to normal upon completion of the test.

The procedures included a section for recommended changes to plant operating procedures and/or the station plant manual that might be generated as a result of the tests. The procedures

also contained a section for remarks related to the tests. A standard cover sheet facilitated formal test procedure review and approval.

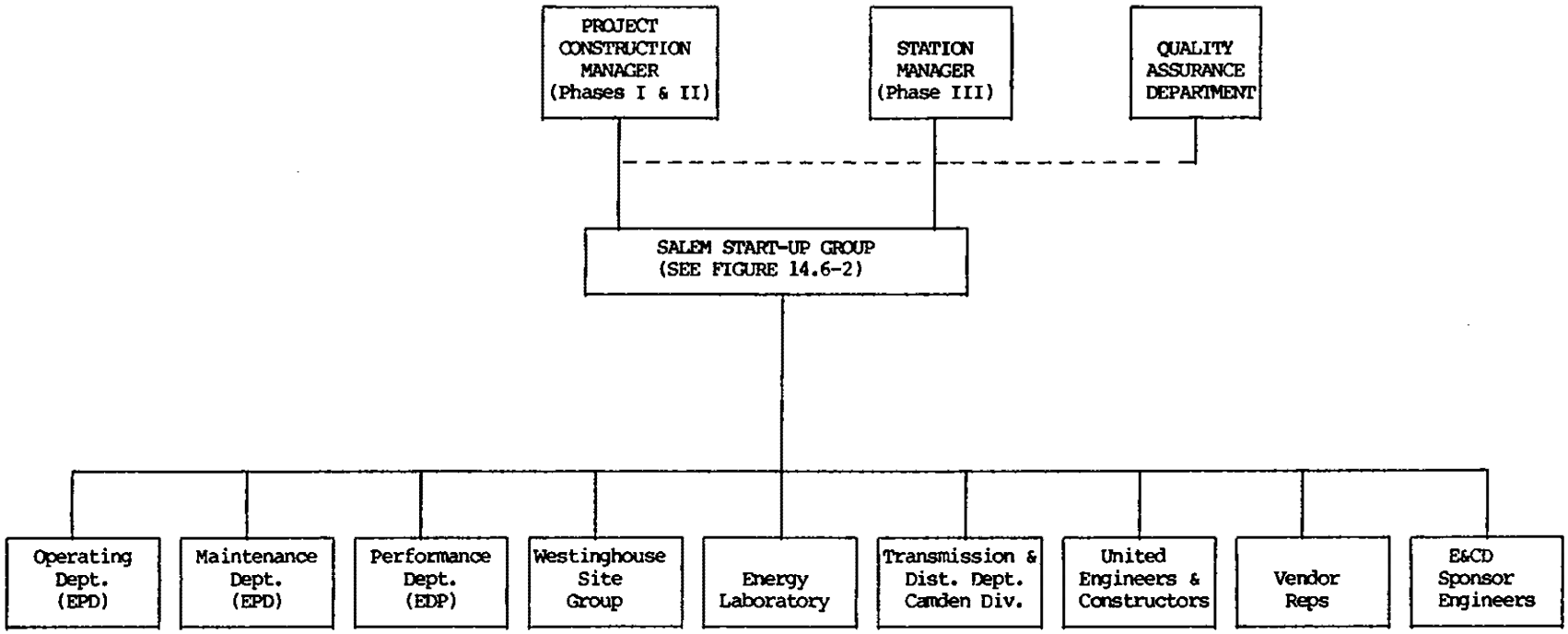
14.6.3 Startup Procedure Changes

Changes to approved startup test procedures were reviewed and approved by the same functional groups involved in the preparation of the original document. A change notice was issued which included the change(s) and reason(s) for the change(s).

On-the-spot changes to startup procedures were made only if the change did not modify the original intent of the procedure. A post-change review, however, had to be subsequently conducted by the same functional groups involved in the approval of the original document.

14.6.4 Startup Test Results

Phase II and III startup procedure results were reviewed and approved by the same functional groups who reviewed and approved the original test procedure. The PSSUG Startup Engineer initiated the review process by ensuring that the procedure had been completed and that any exceptions or comments listed by the test engineer who conducted that test were clearly explained. Copies of the final approved test procedure and results were forwarded to the proper quality organization for record retention.



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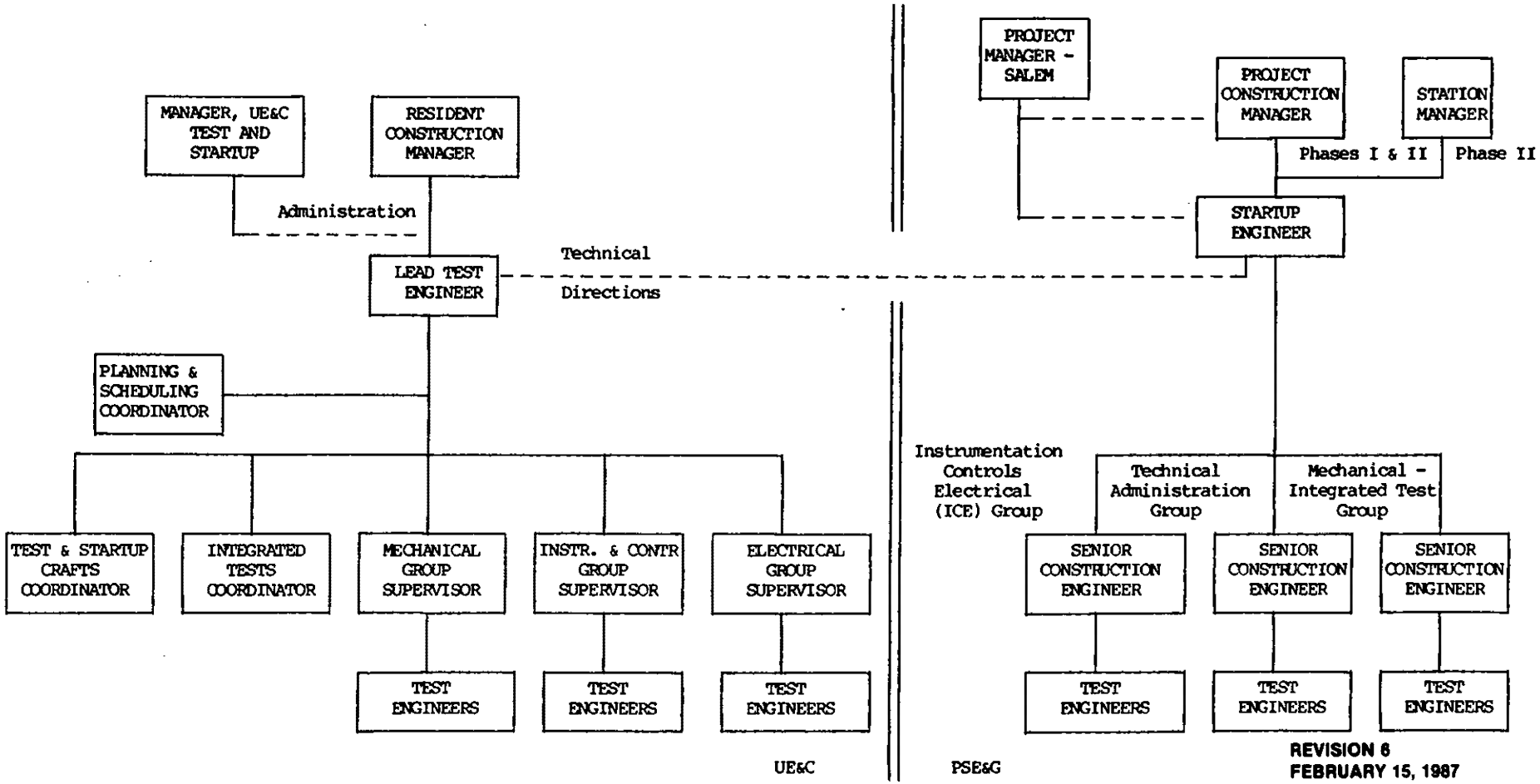
PUBLIC SERVICE ELECTRIC AND GAS COMPANY SALEM NUCLEAR GENERATING STATION	Startup Program Organization - No. 2 Unit
	Updated FSAR — Figure 14.6-1



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PUBLIC SERVICE ELECTRIC AND GAS COMPANY SALEM NUCLEAR GENERATING STATION	Typical Startup Group Organization - No. 2 Unit
	Updated FSAR — Figure 14.6-2